10.14 Real characters and real zeros

The foregoing sections describe the real characters and the primitive characters modulo a prime power. We now use these results to study the series

$$L(\sigma, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^{\sigma}}, \qquad \sigma > 0,$$

where χ is a real nonprincipal character. By the convergence theorem (Theorem 6.17), this series converges for every real $\sigma > 0$, and in Section 6.10 we proved that $L(1, \chi) \neq 0$.

A real zero near 1 is called an exceptional zero, or a Landau–Siegel zero. Siegel's theorem of 1935 implies that if χ is a real primitive character mod q and $L(\beta, \chi) = 0$, where $0 < \beta < 1$, then, for every $\epsilon > 0$,

$$1 - \beta > c(\epsilon) q^{-\epsilon},$$

where $c(\epsilon) > 0$; his proof gives no method for calculating this constant, so the estimate is ineffective. Moreover, for fixed $\epsilon > 0$, $q^{-\epsilon}$ becomes much smaller than $1/\log q$ as q grows. Thus Siegel's theorem does not give a calculable constant $c > 0$ for which $1 - \beta \geq c/\log q$. Theorem 10.16 gives such a result with $c = 10^{-6}$. Such a zero affects estimates for primes in arithmetic progressions, is closely related to class numbers of quadratic fields, and produces the Deuring–Heilbronn phenomenon. Unlike Siegel's class-number method, the proof below derives incompatible upper and lower estimates for the weighted primes with $\chi(p) = -1$. Remarkably enough, the first uniform exclusion of Landau–Siegel zeros was obtained by an AI system [1]. The argument presented here follows the methods of [1, 2, 3]; related accounts include [4, 5, 6, 7, 8]. The essential ideas of the proof can be explained by elementary methods accessible to readers familiar with the material of this book. We prove the following effective estimate.

Theorem 10.16 *Let χ be a real primitive character mod q, where $q \geq 3$. If $0 < \beta < 1$ and $L(\beta, \chi) = 0$, then*

$$(1 - \beta) \log q \geq 10^{-6}. \tag{26}$$

For $X \geq 2$, let

$$S_{-}(X) = \sum_{\substack{p \leq X \\ \chi(p) = -1}} \frac{\log p}{p}. \tag{27}$$

The proof of Theorem 10.16 is based on two estimates for this sum. In Sections 10.17 and 10.18 we obtain an upper estimate by constructing a nonzero integer D with prescribed prime divisors. In Sections 10.19 and 10.20 a zero β near 1 gives a lower estimate incompatible with the first. We complete the proof in Section 10.21. All constants are explicit.

We first put a real primitive character into a form suitable for the congruence argument. Here and below $(d \mid p)$ denotes the Legendre symbol of Chapter 9. The purpose of d is to turn the condition $\chi(p) = -1$ into the elementary assertion that d is a quadratic nonresidue modulo p. This will make the ring generated by $\sqrt{d}$ useful modulo such primes.

Theorem 10.17 *For the character in Theorem 10.16 there is a nonzero squarefree integer $d \neq 1$, with $|d| \leq q$, such that*

$$\chi(p) = (d \mid p) \qquad (p \text{ odd prime}, \ p \nmid q).$$

For example, the primitive real characters modulo 8 correspond to $d = 2$ and $d = -2$. For the quadratic character modulo 5, one may take $d = 5$.

Proof. We first determine the form of q. Suppose first that $q = 2^\alpha$. Since $q \ge 3$, we have $\alpha \ge 2$. For $\alpha = 2$, the discussion preceding Theorem 10.15 shows that the unique nonprincipal character mod 4 is primitive. If $\alpha \ge 3$, Theorem 10.13 shows that a real character $\chi_{a,c}$ must have

$$c = \frac{\varphi(2^\alpha)}{2} = 2^{\alpha-2} \quad \text{or} \quad c = \frac{\varphi(2^\alpha)}{4} = 2^{\alpha-3},$$

whereas Theorem 10.15 says that $\chi_{a,c}$ is primitive if and only if c is odd. When $\alpha = 3$, the two values of c are 2 and 1, so the latter gives the primitive real characters mod 8. When $\alpha \ge 4$, both values are even, so no real character mod 2^α is primitive. Hence $q = 4$ or $q = 8$. If q is not a power of 2, its odd part is squarefree and its 2-part is 1, 4, or 8. Here is a hint for the verification left to the reader in Exercise 20. Factor the character into its prime-power components as at the end of Section 10.13. For an odd prime-power component, Theorems 10.12 and 10.14 rule out an exponent at least 2; for the 2-power component, use Theorems 10.13 and 10.15, remembering that the only character mod 2 is induced from mod 1. Thus, in all cases,

$$q = 2^a m, \qquad a = 0, 2, \text{ or } 3, \qquad m \text{ odd and squarefree.}$$

For each odd prime $r \mid m$, the factorization and primitivity criterion at the end of Section 10.13 show that the corresponding component modulo r is primitive. Theorem 10.12 then identifies it as the quadratic character. Put $r^* = (-1)^{(r-1)/2} r$. Quadratic reciprocity and its supplementary law (Theorems 9.8 and 9.4) give

$$(r^* \mid p) = (p \mid r) \qquad (p \ne r,\ p \text{ odd prime}).$$

For the factor 2^a, the possible primitive real characters, on odd integers n, are

$$(-1)^{(n-1)/2}, \qquad (-1)^{(n^2-1)/8}, \qquad (-1)^{(n-1)/2+(n^2-1)/8},$$

with conductors 4, 8, and 8, respectively. These are the characters constructed in Section 10.11. By Theorems 9.4 and 9.5 their values at an odd prime are $(-4 \mid p)$, $(8 \mid p)$, $(-8 \mid p)$, respectively. We choose $D_2 = 1, -4, 8$, or -8, as required, and let

$$D_0 = D_2 \prod_{r \mid m} r^*.$$

Then $|D_0| = q$ and $\chi(p) = (D_0 \mid p)$ for every odd prime $p \nmid q$. We take $d = D_0$ if $a = 0$, and $d = D_0/4$ otherwise. Since 4 is a square modulo every odd prime, this does not change the Legendre symbol. This d is squarefree and $|d| \le q$. If $m > 1$, an odd prime divides d; if $m = 1$, its possible values are $-1, 2, -2$. Hence $d \ne 1$, as required. □

10.15 Elementary estimates with constants

We shall use the functions $\psi(x)$ and $\vartheta(x)$ of Section 4.2 and the Mangoldt function $\Lambda(n)$ of Section 2.8. Theorem 4.10 supplies the estimates $\vartheta(x) = O(x)$ and $\sum_{p \le x} (\log p)/p = \log x + O(1)$. To obtain the numerical constant in (26), we need explicit versions of these estimates. The following elementary bounds will suffice.

Theorem 10.18 *For every real $x \ge 2$,*

$$\vartheta(x) \le \psi(x) \le 4x, \qquad \log x - 5 \le \sum_{p \le x} \frac{\log p}{p} \le \log x + 5. \tag{28}$$

Proof. Let $v_p(N)$ denote the exponent of p in the nonzero integer N. Legendre's identity (Theorem 3.14) gives

$$v_p\binom{2m}{m} = \sum_{j\ge 1} \left(\left[2m/p^j\right] - 2\left[m/p^j\right] \right) \qquad (m \ge 1).$$

Every summand is 0 or 1, and it is 1 if $m < p^j \le 2m$. Therefore

$$\psi(2m) - \psi(m) \le \log \binom{2m}{m} \le 2m \log 2.$$

The last inequality follows from $\binom{2m}{m} \le \sum_{j=0}^{2m} \binom{2m}{j} = 2^{2m}$. We prove $\psi(n) \le 4n$ by induction. It holds at $n = 1$. For $n \ge 2$, let $m = [(n+1)/2] < n$. Then

$$\psi(n) \le \psi(2m) \le \psi(m) + 2m \log 2 < 6m \le 4n.$$

This proves $\psi(n) \le 4n$ for integral n. If x is real, apply this result with $n = [x]$.

The factorial identity (Theorem 3.12) gives, for an integer $n \ge 1$,

$$\log(n!) = \sum_{j \le n} \Lambda(j) \, [n/j] \, .$$

Since $0 \le n/j - [n/j] < 1$, it follows that

$$\frac{\log(n!)}{n} \le \sum_{j \le n} \frac{\Lambda(j)}{j} \le \frac{\log(n!)}{n} + \frac{\psi(n)}{n}.$$

The integral comparison $n \log n - n + 1 \le \log(n!) \le n \log n$ now gives

$$\log n - 1 \le \sum_{j \le n} \frac{\Lambda(j)}{j} \le \log n + 4.$$

The prime powers of exponent at least 2 contribute at most

$$\begin{aligned} \sum_p \sum_{a \ge 2} \frac{\log p}{p^a} = \sum_p \frac{\log p}{p(p-1)} &\le 2 \sum_{j=2}^{\infty} \frac{\log j}{j^2} \\ &\le 2\left(\frac{\log 2}{4} + \int_2^{\infty} \frac{\log t}{t^2} \, dt \right) < 3. \end{aligned}$$

The integral comparison is valid since $(\log t)/t^2$ decreases for $t \ge 2$. Consequently the prime sum at an integer n lies between $\log n - 4$ and $\log n + 4$. For real $x \ge 2$, we take $n = [x]$. Then $x/2 \le n \le x$, so $\log x - 1 < \log n \le \log x$. This proves (28). □

10.16 The Frobenius congruence and root counting

For a prime with $\chi(p) = -1$, the integer d is a quadratic nonresidue modulo p. Consequently the two-coordinate ring used below becomes a field modulo p, and the Frobenius map sends each element to its conjugate. A differentiated form of this identity will later force powers of p to divide the rows of an integer matrix.

Fix the integer d of Theorem 10.17, and write

$$\mathcal{O} = \mathbb{Z}[\sqrt{d}] = \{a + b\sqrt{d} : a, b \in \mathbb{Z}\}.$$

The coordinates a, b are unique. If $\alpha = a + b\sqrt{d}$, write $\alpha' = a - b\sqrt{d}$. For negative d, the square root is taken in the complex numbers. Congruence modulo $p\mathcal{O}$ means congruence of both integer

coordinates modulo p. We use $\mathcal{O}$ only as a ring of pairs of integers; no theory of algebraic integers or ideals is needed. Modulo $p\mathcal{O}$, it has the p^2 classes $a + b\theta$, where a, b are integers mod p and $\theta^2 = d$.

If $F(z, w)$ is a polynomial, $F^{(j)}(z, w)$ will denote its jth derivative with respect to z, with w held fixed.

Lemma 10.19 *If p is odd and $\chi(p) = -1$, the residue classes modulo $p\mathcal{O}$ form a field of characteristic p.*

Proof. Every residue class has the form $a + b\theta$, where a, b are integers mod p and $\theta^2 = d$. A nonzero class has inverse

$$(a + b\theta)^{-1} = \frac{a - b\theta}{a^2 - db^2}.$$

The denominator does not vanish. For if $b \neq 0$, its vanishing would give $d = (a/b)^2$, contrary to $(d \mid p) = -1$; and if $b = 0$, then $a \neq 0$. Hence every nonzero class has an inverse, as asserted. □

Theorem 10.20 *If p is odd and $\chi(p) = -1$, then*

$$\alpha^p \equiv \alpha' \pmod{p\mathcal{O}} \qquad (\alpha \in \mathcal{O}). \tag{29}$$

Moreover, for every $\alpha \in \mathcal{O}$, if $G(z, w)$ has integer coefficients and h, j are nonnegative integers, then

$$\left(G(z, w)(z^p - w)^h\right)^{(j)}\big|_{(\alpha, \alpha')} \in p^h\mathcal{O}. \tag{30}$$

Proof. Since $\binom{p}{j}$ is divisible by p for $0 < j < p$, the binomial theorem, Fermat's theorem, and Euler's criterion (Theorems 5.19 and 9.2) give

$$(a + b\sqrt{d})^p \equiv a + b\,d^{(p-1)/2}\sqrt{d} \equiv a - b\sqrt{d} \pmod{p\mathcal{O}}.$$

This proves (29). Also

$$(z^p - w)^{(j)} = \begin{cases} p(p-1)\cdots(p-j+1)z^{p-j}, & 1 \leq j \leq p, \\ 0, & j > p. \end{cases}$$

At (α, α'), every derivative of $z^p - w$, including the derivative of order zero, is therefore in $p\mathcal{O}$. When we apply the product rule to $G(z, w)(z^p - w)^h$, every term contains a derivative of each of the h factors $z^p - w$, and hence contains h factors of p. This proves (30). □

Note. The derivatives in (30) are ordinary derivatives; they are not divided by $j!$. Also, j need not be less than p.

Lemma 10.21 *Let the coefficients of $F(z, w)$ belong to a field K of prime characteristic P, and let $1 \leq m \leq P$. Suppose the degree of F in z is less than P, and its degree in w is less than m. If*

$$F^{(j)}(a, a^P) = 0 \qquad (0 \leq j < m)$$

at P distinct elements $a \in K$, then $F = 0$.

Proof. Suppose $F \neq 0$, and let $g(z) = F(z, z^P)$. Distinct monomials $z^r w^s$ give distinct exponents $r + Ps$, since $0 \leq r < P$. Indeed, an equality $r_1 + Ps_1 = r_2 + Ps_2$ first gives $r_1 = r_2$ modulo P, and then gives $s_1 = s_2$. Hence $g \neq 0$, and $\deg g \leq Pm - 1$. The derivative of z^P is zero in characteristic P, so

$$g^{(j)}(z) = F^{(j)}(z, z^P).$$

If $g(z) = \sum c_j (z-a)^j$, then $g^{(j)}(a) = j! c_j$. Since $j < m \le P$, none of $0!, 1!, \dots, (m-1)!$ vanishes in K. The hypotheses therefore imply that $(z-a)^m$ divides $g(z)$. The factor theorem, used in the proof of Lagrange's theorem (Theorem 5.21), now shows that the P distinct values of a give $\deg g \ge Pm$, a contradiction. □

10.17 A determinant from triangular lattice points

We now prove the arithmetic estimate needed for Theorem 10.16.

Theorem 10.22 *For $X \ge 2$ and every integer $k \ge 3$,*

$$S_-(X) \le \left(\frac{1}{2} + \frac{7}{2k}\right) \log X + \log q + \log k + 23. \tag{31}$$

The construction has two distinct uses. The largest prime P will show that the rows of a matrix are independent, and hence that one integer minor D is nonzero with $P \nmid D$. Every smaller prime $p < P$ for which $\chi(p) = -1$ will then force a large power of p to divide the same integer D. Comparing these divisibilities with an upper bound for $|D|$ will give the desired estimate for $S_-(X)$.

Let P be the largest odd prime not exceeding X for which $\chi(P) = -1$. Thus the capital letter P denotes the prime used to prove that $D \ne 0$, whereas a lower-case p will later denote one of the smaller prime divisors forced upon D. The construction is needed only when $P \ge 5$ and $k \le P$; throughout this section we assume these inequalities and let $n = P - 1$.

The weight $r + ks$ is chosen with the later row operation in mind. Replacing one factor z^p by w changes an exponent pair by $(-p, 1)$. When $p \ge k$, this cannot increase $r + ks$, so the following triangular set remains closed under that operation.

Let $\mathscr{T}$ be the set of pairs of nonnegative integers satisfying

$$\mathscr{T} = \{(r,s) : r + ks \le n\},$$

and write

$$t = |\mathscr{T}|, \quad U = \sum_{\mathscr{T}} r, \quad V = \sum_{\mathscr{T}} s, \quad J = [n/k], \quad m = J + 1.$$

Thus t is the number of allowed monomials, U and V are their total z- and w-degrees, J is the largest possible w-degree, and $m = J + 1$ is the number of derivative orders $0, \dots, J$ used at each point. Since $n \ge 4$, we have $U > 0$. Every polynomial of the form

$$\sum_{(r,s)\in\mathscr{T}} c_{rs} z^r w^s$$

has degree less than P in z and less than m in w. We shall use the inequalities

$$t \le P^2, \qquad nt \le 3U, \qquad kV \le U. \tag{32}$$

For a fixed s, the reflection $r \mapsto n - ks - r$ permutes the integers in the interval $0 \le r \le n - ks$; hence

$$2 \sum_{r=0}^{n-ks} r = (n - ks)(n - ks + 1).$$

Summing over s, we obtain

$$2U = \sum_{s=0}^{J} (n - ks)(n - ks + 1) = nt - kV,$$

and therefore $2U + kV = nt$. To prove $kV \le U$, we write $r = ka + b$, where $0 \le b < k$. For each b, the allowed pairs (a, s) form the triangle $a + s \le [(n - b)/k]$, which is unchanged when a and s are interchanged. Hence

$$U = k \sum a + \sum b \ge k \sum s = kV,$$

and consequently $nt = 2U + kV \le 3U$. Also $\mathcal{T} \subseteq \{0, \dots, P-1\}^2$, so $t \le P^2$.

Let $N = 1 + \left[\sqrt{P}\right]$, and select P distinct pairs from the N^2 pairs of integers $0 \le u, v < N$. To each chosen pair we associate $\alpha = u + v\sqrt{d}$ and its conjugate α'. Their reductions modulo $P\mathcal{O}$ are distinct, since $N \le P$, and Theorem 10.20 gives $\alpha' = \alpha^P$ after reduction. Moreover,

$$|\alpha|, |\alpha'| < H, \qquad H = 3\sqrt{qP},$$

because $N \le \sqrt{P} + 1 \le (3/2)\sqrt{P}$ and $N(1 + \sqrt{|d|}) \le 2N\sqrt{q}$.

The matrix records the first m derivatives of every allowed monomial at the chosen conjugate pairs. More precisely, its rows are indexed by the pairs $(r, s) \in \mathcal{T}$. Its columns are indexed by the pairs (α, j), where α runs through the P chosen points and $0 \le j \le J$. In row (r, s) and column (α, j) we place

$$(z^r w^s)^{(j)}\big|_{(\alpha, \alpha')}, \tag{33}$$

Each entry belongs to $\mathcal{O}$. Replacing an entry $a + b\sqrt{d}$ by its two integer coordinates a, b, we obtain an integer matrix with t rows and $2Pm$ columns.

Lemma 10.23 *This integer matrix has a square minor D of order t such that*

$$D \ne 0 \qquad \textit{and} \qquad P \nmid D.$$

Proof. Suppose the rows are linearly dependent modulo P. Then there are coefficients c_{rs}, not all zero, such that the polynomial

$$F(z, w) = \sum_{(r,s) \in \mathcal{T}} c_{rs} z^r w^s,$$

has its first m derivative values equal to zero at the P distinct pairs (a, a^P) in the field $\mathcal{O}/P\mathcal{O}$ of Lemma 10.19. Since $m = [(P-1)/k] + 1 \le P$, Lemma 10.21 gives $F = 0$. Hence every $c_{rs} = 0$, a contradiction. Therefore the rows are independent modulo P, and some t columns determine a square minor which is nonzero modulo P. Its determinant is the required integer D. □

10.18 An upper bound for $S_-(X)$

Proof of Theorem 10.22. Let P have the meaning assigned in the foregoing section. If there is no such odd prime, or if $P = 3$, then

$$S_-(X) \le \frac{\log 2}{2} + \frac{\log 3}{3} < 1,$$

and the theorem follows. If $k > P$, all the primes occurring in $S_-(X)$ are less than k, and Theorem 10.18 gives $S_-(X) \le \log k + 5$. We may therefore assume that $P \ge 5$ and $k \le P$, and we use the determinant D of Lemma 10.23.

We now estimate this same integer D in two ways. Bounds for its entries give an upper bound for $\log |D|$. On the other hand, each smaller prime p with $\chi(p) = -1$ divides D to a known order. Prime factorization then gives a lower bound for $\log |D|$.

Size of the determinant. A nonzero entry in row (r,s) of (33) is $(r)_j \alpha^{r-j}(\alpha')^s$, where $(r)_j = r(r-1)\cdots(r-j+1)$. Since $r \le n$, $j \le J$, and $H \ge 1$, this entry and its conjugate have modulus at most $n^J H^{r+s}$. If $\gamma = a + b\sqrt{d}$ and γ' both have modulus at most M, then

$$|a| = \left|\frac{\gamma+\gamma'}{2}\right| \le M, \qquad |b| = \left|\frac{\gamma-\gamma'}{2\sqrt{d}}\right| \le \frac{M}{\sqrt{|d|}} \le M.$$

The same bound therefore holds for both integer coordinates. Index the rows by $i = 1,\dots,t$, and let (r_i,s_i) be the pair in row i. If $a_{i\ell}$ is any integer entry in that row, the preceding estimate gives

$$|a_{i\ell}| \le n^J H^{r_i+s_i}.$$

For the t columns defining D, we have $\sum_i r_i = U$ and $\sum_i s_i = V$. The Leibniz formula now gives

$$\begin{aligned} |D| &= \left|\sum_{\pi\in\mathfrak{S}_t} \operatorname{sgn}(\pi) \prod_{i=1}^{t} a_{i,\pi(i)}\right| \\ &\le \sum_{\pi\in\mathfrak{S}_t} \prod_{i=1}^{t} |a_{i,\pi(i)}| \le t! \prod_{i=1}^{t} n^J H^{r_i+s_i} = t!\, n^{Jt} H^{U+V}. \end{aligned}$$

Using $\log(t!) \le t\log t$, $J \le n/k$, and (32), we obtain

$$\begin{aligned} \frac{\log|D|}{U} &\le \left(1+\frac{1}{k}\right)\log H + \frac{3}{k}\log n + \frac{6\log P}{P-1} \\ &\le \left(\frac{1}{2}+\frac{7}{2k}\right)\log P + \log q + 5. \end{aligned} \tag{34}$$

For the last inequality we used $\log H = \frac{1}{2}\log P + \frac{1}{2}\log q + \log 3$, $(1+1/k)\log 3 < 2$, and $\log P/(P-1) \le 1/2$. The coefficient of $\log q$ is $\frac{1}{2}(1+1/k) \le 2/3$.

Divisibility by a smaller bad prime. Let p be a prime such that $k \le p < P$ and $\chi(p) = -1$. For each row we write $r = ph + c$, where $h = [r/p]$ and $0 \le c < p$, and replace its monomial by

$$z^c(z^p - w)^h w^s = z^r w^s + \sum_{i=1}^{h} (-1)^i \binom{h}{i} z^{r-pi} w^{s+i}. \tag{35}$$

For example, if $r = 7$ and $p = 3$, the row change is

$$z^7 w^s \longmapsto z(z^3-w)^2 w^s = z^7 w^s - 2z^4 w^{s+1} + z w^{s+2}.$$

The two additional monomials have smaller z-degree; when $\chi(3) = -1$, every evaluated coordinate in the new row is divisible by 3^2. All the new exponent pairs remain in $\mathscr{T}$, since

$$(r-pi) + k(s+i) = r + ks - i(p-k) \le n,$$

and they have smaller z-degree. Hence, when the rows are ordered by increasing r, the replacements are given by an integer lower triangular matrix with diagonal entries 1 and leave the fixed minor D unchanged. By (30), both integer coordinates in the new row are divisible by p^h. Factoring these powers from the rows gives

$$v_p(D) \ge \sum_{(r,s)\in\mathscr{T}} [r/p] \ge U/p - t.$$

Comparison of the two estimates. Since D is a nonzero integer, unique factorization gives

$$\log|D| = \sum_{\ell \mid D} v_\ell(D)\log\ell,$$

where the sum is over the prime divisors of D. Retaining only the primes $k \le p < P$ for which $\chi(p) = -1$, and then using Theorem 10.18, we obtain

$$\begin{aligned}\log|D| &\ge U \sum_{\substack{k\le p<P\\ \chi(p)=-1}} \frac{\log p}{p} - t\vartheta(P-1)\\ &\ge U \sum_{\substack{k\le p<P\\ \chi(p)=-1}} \frac{\log p}{p} - 12U,\end{aligned}$$

where we used $(P-1)t \le 3U$. The primes below k, including the prime 2, contribute at most $\log k + 5$, and the prime P contributes at most 1. By the choice of P, we have $S_-(X) = S_-(P)$, and hence

$$S_-(X) \le \frac{\log|D|}{U} + 12 + \log k + 5 + 1.$$

Together with (34), this proves (31) and completes the proof. □

10.19 A lower bound for $S_-(X)$

We return to the nonnegative divisor sum of Theorem 6.19. Let

$$A(n) = \sum_{a\mid n} \chi(a), \qquad B_\beta(x) = \sum_{n\le x} \frac{A(n)}{n^\beta}, \qquad 0 < \beta < 1.$$

By that theorem, $A(n) \ge 0$ and $A(1) = 1$; consequently $B_\beta(x) \ge 1$ for $x \ge 1$. We shall use also

$$A(pn) \ge A(n) \qquad \text{if } \chi(p) \ge 0. \tag{36}$$

By Theorem 2.14, A is multiplicative. The prime-power values in the proof of Theorem 6.19 are $A(p^e) = e+1$ if $\chi(p) = 1$ and $A(p^e) = 1$ if $\chi(p) = 0$. They are nondecreasing in e, which proves (36), including the case $p \mid n$.

The following finite-sum inequality relates these auxiliary sums to the prime weight S_-.

Lemma 10.24 *If $0 < \beta < 1$, $X \ge 2$, and $Y > 1$, then*

$$S_-(X) \ge 2\log X - \frac{B_\beta(XY)}{B_\beta(Y)} \log(XY) - 5. \tag{37}$$

Proof. Let $T = XY$, and denote by $S_{\ge 0}(X)$ the sum of $(\log p)/p$ over the primes $p \le X$ with $\chi(p) \ge 0$. It includes the primes dividing q, and is complementary to $S_-(X)$ in the full prime sum. For $p \le X$, $T/p \ge Y$. Since $A(n) \ge 0$, the function $B_\beta(x)$ is nondecreasing, and hence $B_\beta(T/p) \ge B_\beta(Y)$. We also have $p^{-\beta} \ge p^{-1}$. Therefore

$$\begin{aligned}B_\beta(Y) S_{\ge 0}(X) &\le \sum_{\substack{p\le X\\ \chi(p)\ge 0}} \frac{\log p}{p^\beta} B_\beta(T/p)\\ &= \sum_{n\le T} \frac{1}{n^\beta} \sum_{\substack{p\mid n,\ p\le X\\ \chi(p)\ge 0}} A(n/p)\log p\\ &\le \sum_{n\le T} \frac{A(n)}{n^\beta} \sum_{p\mid n} \log p \le \sum_{n\le T} \frac{A(n)}{n^\beta} \log n.\end{aligned}$$

Here we used (36) and the fact that the product of the distinct primes dividing n divides n.

We separate the terms with $n \le Y$ from those with $Y < n \le T$. Using $\log n \le \log Y$ in the first sum and $\log n \le \log T$ in the second, we obtain

$$\sum_{n \le T} \frac{A(n)}{n^\beta} \log n \le B_\beta(Y) \log Y + (B_\beta(T) - B_\beta(Y)) \log T$$
$$= B_\beta(T) \log T - B_\beta(Y) \log X.$$

This is also an upper bound for $B_\beta(Y) S_{\ge 0}(X)$. After division by $B_\beta(Y) > 0$, we subtract this inequality from the lower bound $\log X - 5$ for the full prime sum in Theorem 10.18, and obtain (37). □

10.20 Weighted divisor sums at a real zero

We next estimate $B_\beta(x)$ when $L(\beta, \chi) = 0$. As in the proof of Theorem 6.20, we use the hyperbola identity. We shall need an error estimate which is uniform for $3/4 \le \beta < 1$. The zero will produce a main term involving $L(1, \chi)$. The chief point is to combine two remaining tails before estimating them, so that their factors $1/(1-\beta)$ cancel.

For $\sigma > 0$, let

$$R_\sigma(u) = \sum_{n > u} \frac{\chi(n)}{n^\sigma}, \qquad L_\sigma(u) = \sum_{n \le u} \frac{\chi(n)}{n^\sigma}.$$

The periodic cancellation in the proof of Theorem 6.17 gives $|C_\chi(t)| \le \varphi(q) \le q$, where $C_\chi(t) = \sum_{n \le t} \chi(n)$. If we apply Abel's identity (Theorem 4.2) and let the upper endpoint tend to infinity, we obtain

$$R_\sigma(u) = -C_\chi(u) u^{-\sigma} + \sigma \int_u^\infty C_\chi(v) v^{-\sigma-1}\, dv, \qquad |R_\sigma(u)| \le 2q u^{-\sigma}. \tag{38}$$

The integral is absolutely convergent. This gives the estimate from Theorem 6.17 which we shall use.

Let $\delta = 1 - \beta$ and $H_\beta(v) = \sum_{n \le v} n^{-\beta}$. The proof of Theorem 3.2(b) gives, more precisely,

$$H_\beta(v) = \frac{v^\delta}{\delta} + \zeta(\beta) + e_\beta(v), \qquad |e_\beta(v)| \le 2v^{-\beta} \quad (v \ge 1). \tag{39}$$

Here $\zeta(\beta)$ denotes the constant occurring in Theorem 3.2(b), not the sum of a convergent infinite series when $0 < \beta < 1$. In fact, if $\{y\} = y - [y]$, that proof gives

$$e_\beta(v) = -\{v\} v^{-\beta} + \beta \int_v^\infty \{y\} y^{-\beta-1}\, dy,$$

and the estimate follows from $0 \le \{y\} < 1$.

Lemma 10.25 *If $3/4 \le \beta < 1$ and $L(\beta, \chi) = 0$, then for $x \ge 1$,*

$$B_\beta(x) = \frac{L(1, \chi)}{\delta} x^\delta + E(x), \qquad |E(x)| \le 10 q x^{-1/4}. \tag{40}$$

Proof. Let $u = \sqrt{x}$. The hyperbola identity (Theorem 3.17), applied to $f(a) = \chi(a) a^{-\beta}$ and $g(b) = b^{-\beta}$, gives

$$B_\beta(x) = \sum_{a \le u} \frac{\chi(a)}{a^\beta} H_\beta(x/a) + \sum_{b \le u} \frac{1}{b^\beta} L_\beta(x/b) - H_\beta(u) L_\beta(u). \tag{41}$$

On inserting (39) into the first and last terms, the two terms $\zeta(\beta) L_\beta(u)$ cancel. Since $\beta + \delta = 1$, we obtain

$$B_\beta(x) = \frac{x^\delta}{\delta} L_1(u) - \frac{u^\delta}{\delta} L_\beta(u) + E_0(x). \tag{42}$$

Here the error term is

$$E_0(x) = \sum_{a \le u} \frac{\chi(a)}{a^\beta} e_\beta(x/a) + \sum_{b \le u} \frac{1}{b^\beta} L_\beta(x/b) - e_\beta(u) L_\beta(u).$$

Since $L(\beta, \chi) = 0$, we have $L_\beta(v) = -R_\beta(v)$, so (38) bounds the three errors by

$$\left| \sum_{a \le u} \frac{\chi(a)}{a^\beta} e_\beta(x/a) \right| \le 2ux^{-\beta} = 2u^{1-2\beta},$$

$$\left| \sum_{b \le u} \frac{1}{b^\beta} L_\beta(x/b) \right| \le 2qux^{-\beta} = 2qu^{1-2\beta},$$

$$|e_\beta(u) L_\beta(u)| \le 4qu^{-2\beta} \le 4qu^{1-2\beta}.$$

Consequently $|E_0(x)| \le (2 + 6q)u^{1-2\beta}$.

The two terms outside $E_0(x)$ must not be estimated separately: each contains the large factor $1/\delta$. Their difference is much smaller. Indeed, $R_\beta(u)$ is obtained from the tail $R_1(u)$ by inserting the weight n^δ, and Abel summation exposes a compensating factor δ. Put $a_n = \chi(n)/n$, so that $R_1(v) = \sum_{n > v} a_n$. For $W > u$, Abel's identity (Theorem 4.2), applied to a_n and the weight v^δ, gives

$$\sum_{u < n \le W} \frac{\chi(n)}{n^\beta} = u^\delta R_1(u) - W^\delta R_1(W) + \delta \int_u^W v^{-\beta} R_1(v)\, dv.$$

By (38), the boundary term at W tends to zero and the integral converges absolutely. Therefore

$$\frac{R_\beta(u) - u^\delta R_1(u)}{\delta} = \int_u^\infty v^{-\beta} R_1(v)\, dv, \qquad \left| \int_u^\infty v^{-\beta} R_1(v)\, dv \right| \le \frac{2q}{\beta} u^{-\beta} \le 3qu^{-\beta}. \tag{43}$$

Using $L_1(u) = L(1, \chi) - R_1(u)$, $L_\beta(u) = -R_\beta(u)$, and $x = u^2$ in (42), we find

$$B_\beta(x) = \frac{L(1, \chi)}{\delta} x^\delta + u^\delta \int_u^\infty v^{-\beta} R_1(v)\, dv + E_0(x).$$

The error is at most

$$(2 + 9q)u^{1-2\beta} \le 10qx^{1/2-\beta} \le 10qx^{-1/4},$$

because $q \ge 3$ and $\beta \ge 3/4$. This proves (40). □

We have the following consequence.

Theorem 10.26 *Under the hypotheses of Lemma 10.25, let $X \ge 2$, $Y > 1$, and $\rho = 10qY^{-1/4}$. Then*

$$S_-(X) \ge 2 \log X - ((1 + \rho)X^\delta + \rho) \log(XY) - 5. \tag{44}$$

Proof. We apply (40) at XY and Y. The coefficient $L(1, \chi)/\delta$ cancels, and we obtain

$$B_\beta(XY) = X^\delta B_\beta(Y) + E(XY) - X^\delta E(Y).$$

The two error terms satisfy $|E(XY)| \le \rho$ and $|E(Y)| \le \rho$. Since $B_\beta(Y) \ge 1$, we have

$$\frac{B_\beta(XY)}{B_\beta(Y)} = X^\delta + \frac{E(XY) - X^\delta E(Y)}{B_\beta(Y)} \le X^\delta + \rho + \rho X^\delta = (1 + \rho)X^\delta + \rho.$$

Since the coefficient of this ratio in (37) is negative, the last inequality gives (44). □

10.21 Exclusion of real zeros near 1

We can now complete the proof of Theorem 10.16. We choose

$$Q = 10q, \qquad k = 64, \qquad X = Q^{128}, \qquad Y = Q^8. \tag{45}$$

These choices have separate purposes. The value $Q = 10q$ ensures $Q \ge 30$. The value $k = 64$ makes the leading coefficient in the determinant bound comfortably smaller than $2/3$. The choice $Y = Q^8$ makes the relative error at most $1/30$. Finally, $X = Q^{128}$ is large enough compared with Y for the lower estimate to exceed $(3/4)\log X$, while the contrary assumption still keeps $(1-\beta)\log X$ small. Thus $XY = Q^{136}$, and $\rho = 10qY^{-1/4} = Q^{-1} \le 1/30$.

Proof of Theorem 10.16. Suppose $L(\beta, \chi) = 0$. If $0 < \beta \le 3/4$, then

$$(1-\beta)\log q \ge \frac{1}{4}\log 3 > 10^{-6},$$

and the theorem follows. We may therefore assume that $3/4 < \beta < 1$. Let $\delta = 1-\beta$, and suppose, to the contrary, that $\delta \le 10^{-6}/\log q$. Let $\ell = \log Q > 3$. By Theorem 10.22,

$$\begin{aligned} S_-(X) &\le \left(\frac{1}{2} + \frac{7}{128}\right) 128\ell + \log q + \log 64 + 23 \\ &< 72\ell + 29 < 82\ell = \frac{41}{64}\log X < \frac{2}{3}\log X. \end{aligned} \tag{46}$$

Here $\log q < \ell$, $\log 64 < 6$, and $29 < 10\ell$.

On the other hand, $\log 10 < 3$ and $\log q > 1$ give

$$0 < \delta \log X \le 128 \cdot 10^{-6}\left(1 + \frac{\log 10}{\log q}\right) < 512 \cdot 10^{-6} < \frac{1}{1000}.$$

For $0 \le v < 1$, comparison of the power series gives $e^v \le (1-v)^{-1}$. Thus $X^\delta \le 1000/999$, and

$$(1+\rho)X^\delta + \rho \le \frac{31}{30}\frac{1000}{999} + \frac{1}{30} = \frac{31999}{29970} < 1.07 < \frac{9}{8}.$$

Theorem 10.26 now gives

$$S_-(X) > 256\ell - \frac{9}{8}136\ell - 5 = 103\ell - 5 > 96\ell = \frac{3}{4}\log X,$$

where $7\ell > 5$ was used. This contradicts (46), and proves the theorem. □

Exercises

21. Let χ be a real nonprincipal character mod m, induced by a primitive character χ^* mod q. Prove that $L(s,\chi)$ and $L(s,\chi^*)$ have the same real zeros for $0 < s < 1$. Deduce from Theorem 10.16 that every such zero β satisfies
$$(1-\beta)\log m \geq 10^{-6}.$$

22. Under the hypotheses of Lemma 10.25, prove that $L(1,\chi) > 0$ and that
$$B_\beta(x) \sim \frac{L(1,\chi)}{1-\beta}\, x^{1-\beta} \qquad (x \to \infty).$$

23. A refinement attributed to K. Barreto improves the dependence on the conductor in Theorem 10.22. Let $P \geq 5$ be the largest prime not exceeding X for which $\chi(P) = -1$, and suppose that $q \leq P^2$.

    (a) Choose P elements $\alpha = x + y\sqrt{d}$ of $\mathcal{O}$, pairwise incongruent modulo $P\mathcal{O}$, such that
    $$|\alpha|, |\alpha'| < 4P^{1/2}q^{1/4}.$$
    Hint. Use a rectangle whose side lengths are close to $P^{1/2}q^{1/4}$ and $P^{1/2}q^{-1/4}$.

    (b) Use these points in the determinant argument of Sections 10.17–10.18 to prove, for every $k \geq 3$,
    $$S_-(P) \leq \left(\frac{1}{2} + \frac{7}{2k}\right)\log P + \frac{k+1}{4k}\log q + \log k + O(1),$$
    where the constant in $O(1)$ is absolute and effective.

    (c) Take $k = \lceil \log P \rceil$, treating the bounded values of P separately. Split into the cases $q \leq P^2$ and $q > P^2$, and prove uniformly for $X \geq 3$ that
    $$S_-(X) \leq \frac{1}{2}\log X + \frac{1}{4}\log q + \log\log X + O(1).$$

Notes and references

The section numbers, theorem numbers, and equation numbers continue Chapter 10 of Apostol [9]. References to earlier numbered theorems and exercises are to that book. These sections are newly prepared text and do not form part of the original 1976 edition.

The terminology and historical summary in Section 10.14 follow the standard account in Chapter 11 of [10]. Siegel's ineffective estimate comes from his 1935 class-number paper [11]. The repulsion associated with an exceptional zero grew out of the class-number work of Deuring [12] and Heilbronn [13]; elementary later treatments include Pintz [14] and Stopple [15].

The nonvanishing principles behind interpolation determinants go back to Masser [16] and Laurent [17]; the related adelic polynomial method appears in Bombieri–Vaaler [18]. The multiplicity principle goes back to Stepanov [19]. The use of an integer minor is attributed to Barreto [4] in the Pescador manuscript. The refinement in Exercise 23 is likewise attributed there to Barreto; both attributions are indirect here.

Bostan–Dumas [20] discusses Wronskians and linear independence, and Rosser–Schoenfeld [21] gives sharper explicit prime estimates. None of these deeper results is required above. This presentation claims no new theorem or priority.

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